ECE 437 Lab 8 — Post Lab: H-Bridge Motor Controller

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Question 1(a): How good is your protractor?

Question. *Report the standard deviation of your 100 stationary readings, in degrees. Compare it with the resolution you predicted in the pre lab, and account for any difference.*

Answer. The standard deviation of the 100 stationary readings of φ (one every 50 ms) is **0.63°**. The pre-lab predicted **0.22°** (1 mg ÷ g·sin 15°).

The difference is accelerometer noise. The pre-lab assumed 1 mg, but the measured noise is **2.0 mg (radial) and 3.0 mg (tangential)**. The measured tilt is β = 14.8°, essentially the assumed 15°. With the measured noise and β, the same formula predicts **0.66°**, which agrees with the measured 0.63°.

Question 3(b): Ringing

Question. *From a continuous accelerometer log of the arm coming to a stop, report the overshoot and the settling time. Then turn that into a rule: what is the fastest rate at which this rig can make trustworthy stop-and-read position measurements? Give a number in measurements per second.*

Answer. Log of a 60-pulse move at 6 V, logged continuously at 400 samples/s:

- **Overshoot: 19.5°** (23 % of an 84.8° move)
- **Settling time: 0.40 s** after the last pulse, until φ stays within ±1.75° of its final value

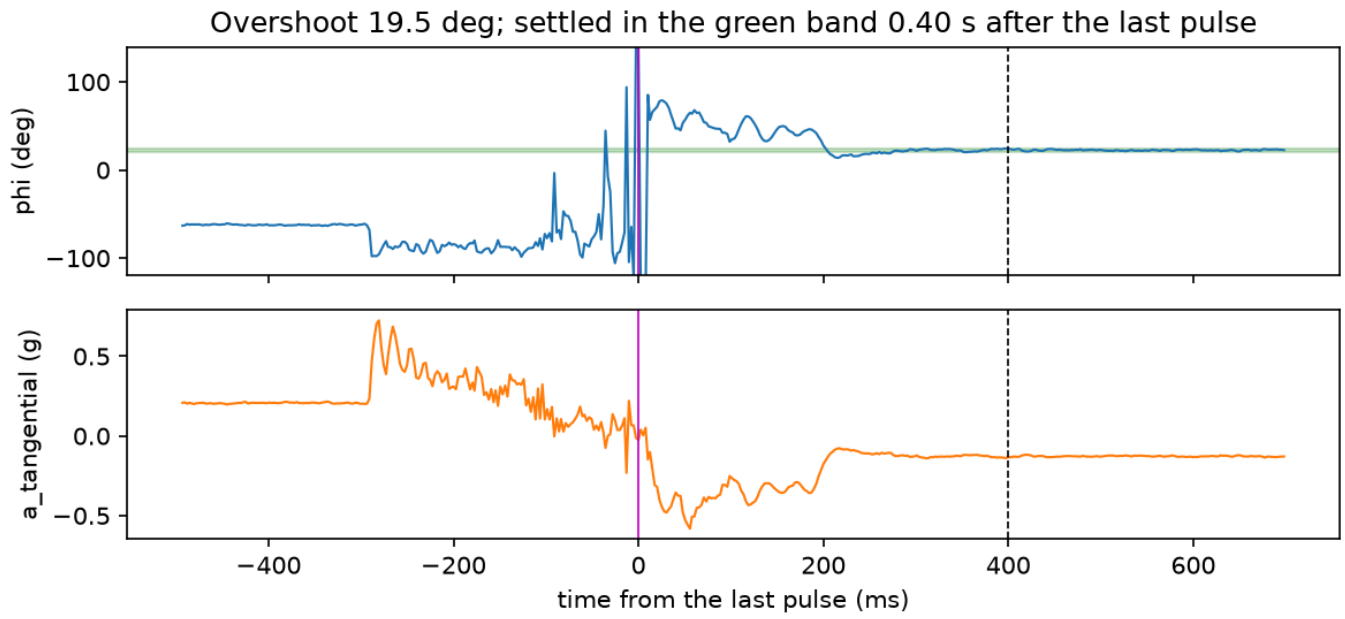


Figure 1. The ringing log around the stop. Magenta = last pulse, dashed = settled.

Rule. Each reading takes the settling time plus a 0.25 s average: 1 / (0.40 s + 0.25 s) ≈ **1.5 measurements per second**. The slowest settling time I measured on another move was 0.62 s, which gives about 1.1 per second, so **about 1 measurement per second** is the safe rate.

Question 4(a): Torque, with no torque sensor

Question. *From a continuous accelerometer log of the arm speeding up from rest, extract the angular acceleration α from the tangential axis. Using the J you computed in the pre lab, report the torque the motor actually produced, in N·m (τ = J·α).*

Answer. From rest, the change in the tangential axis is Δa_t = α·r, so α = Δa_t·g / r with r = 0.10 m. I averaged α over the first 40 ms after GO, before the arm has turned enough for gravity to change a_t. The pre-lab gives J = 0.3 kg × (0.28 m)^2 / 12 = 1.96 × 10^{-3} kg·m^2.

Log (6 V, from rest)	α	τ = J·α
ring_20261103_154409	29.9 rad/s^2	0.059 N·m
6V_20261103_154455	24.9 rad/s^2	0.049 N·m

τ ≈ 0.054 ± 0.005 N·m at 6 V.

Question 4(c): Torque at two voltages

Question. *Repeat 4(a) at two supply voltages. Report both torques and say whether the difference is what you expect, and why.*

Answer. Same move from rest at −62°, 60 pulses, 50 % duty:

Supply	α	τ
6.0 V	24.9 rad/s^2	**0.049 N·m**
4.0 V	14.5 rad/s^2	**0.028 N·m**

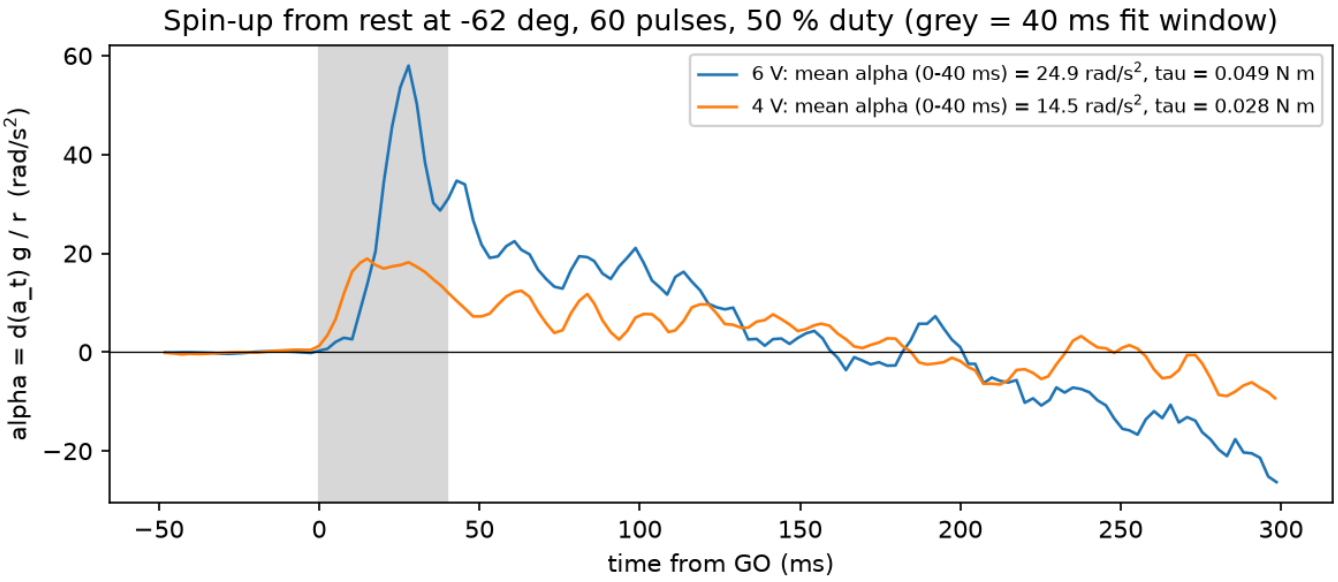


Figure 2. α during the two spin-ups. Grey = the 40 ms averaging window.

Yes, the difference is roughly what I expect. From rest there is no back-EMF, so the current is V/R and the motor's torque should scale with voltage: 6/4 = **1.5×**. The measured ratio is **1.7×**. That is a little higher because the accelerometer sees the **net** torque. A fixed load (gearbox friction and gravity on the tilted arm) is subtracted from both, so it takes a bigger share at 4 V. Solving τ = kV − τ_load with the two points gives τ_load ≈ 0.014 N·m.